1 Preview

Art is the lie that tells us the truth.
—*Pablo Picasso*

All models are wrong but some are useful.
—*George E. P. Box*

Pablo Picasso was a contemporary of George Box's, a statistician who had an enormous impact on his field, writing influential papers well into his 90s. Both men sought to express truth about nature, but they used tools that were dramatically different—Picasso brushing strokes on canvas, and Box writing equations on paper. Given the different ways they worked, it is remarkable that Box and Picasso made such similar statements about the central importance of abstraction to insight. Abstraction plays a role in all creative human enterprise—in art, music, literature, engineering, and science. We create abstractions because they allow us to focus on the most important elements of a problem, those relevant to the objectives of our work, without being distracted by elements that are not relevant.

Scientific models are, above all else, abstractions. They are statements about the operation of nature that purposefully omit many details and thus achieve insight that would otherwise be discursively obscured. They provide unambiguous statements of what we believe is important. A key principle in modeling and statistics—in science for that matter—is the need to reduce the dimensions of a problem. A data set may contain a thousand observations. By reducing its dimensions to a model with a few parameters, we are able to gain understanding.

However, because models are abstractions and reduce the dimensions of a problem, we must deal with the elements we choose to omit. These elements create uncertainty in the predictions of models, so it follows that assessing uncertainty is fundamental to science. Scientists, journalists, logicians, and attorneys alike can rightly claim to make statements based on evidence, but only scientific statements include evidence tempered by uncertainty quantified. We know what is certain only to the extent that we can say, with confidence, what is uncertain. Sharpening our thinking about uncertainty and learning how to estimate it properly is a main theme of this book.

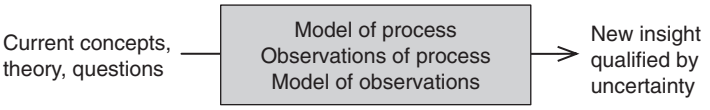


Figure 1.1.1. The fundamental challenge in ecological research is to establish a credible line of inference extending from concepts and theory to new insight tempered by uncertainty.

Your science will have impact to the extent that you are able to ask important questions and provide compelling answers to them. Doing so depends on establishing a line of inference that extends from current thinking, theory, and questions to new insight qualified by uncertainty (fig. 1.1.1). This book offers a highly general, flexible approach to establishing this line of inference. We cannot help you pose novel, interesting questions, but we can teach an approach to inference applicable to an enormous range of research problems, an approach that can be understood from first principles and that can be unambiguously communicated to other scientists, managers, and policy makers. We emphasize that understanding the principles of this framework allows you to customize your analyses to accommodate the inevitable idiosyncrasies of specific problems in research.

We sketch that framework in this chapter to give a general sense of where this book is headed, a preview we use to motivate the development of concepts and principles in the chapters that follow. There should be details of our approach that are unfamiliar, otherwise you probably don't need this book. Soon enough, we will explain those details fully. For now, we offer a somewhat abstract overview followed by a concrete example as an enticement to read on. It will be rewarding to return to this section after you have worked through the book. We hope you will be pleasantly surprised by your increased understanding of our small preview. The only part of this chapter that is essential for the remainder of the book is understanding our notation, which we describe in section 1.1.1.

1.1 A Line of Inference for Ecology

Virtually all research problems in ecology share a set of features. We want to understand how the state of an ecological system changes over time, across space, or among individuals. We seek to understand why those changes occur. Our understanding usually depends on a sample drawn from all possible instances of the state because we want to make statements about a system that is too large to observe fully. The observations in that sample

are often related imperfectly to the true state. In the subsections that follow, we lay out an approach first described by Berliner (1996)[1] for modeling the imperfect observations that arise from a process we want to understand. It does not apply to all research problems, but it is sufficiently general and flexible that it applies to most.

1.1.1 Some Notation

Before we proceed, we must introduce some notation. Boldface lowercase letters will indicate vectors (e.g., $\boldsymbol{\theta}$, $\mathbf{a}$), and lightface lowercase letters, scalars (θ, a).[2] Bold capital letters will be used for matrices (e.g., $\mathbf{A}$). The symbol $\boldsymbol{\theta}$ will indicate a vector of parameters, and, of course, θ will indicate a single parameter.[3] The letter $\mathbf{y}$ will indicate a vector of data, $\mathbf{Y}$ a matrix, and y or y_i a single observation. Corresponding notation using $\mathbf{x}$, x, and x_i will be used for predictor variables, also called covariates. The notation $[a|b, c]$ will be used for the probability distribution of the random variable a conditional on the parameters b and c.[4] Deterministic models will be denoted by $g()$ with arguments necessary for the model within the parentheses. Notation will be added as needed, in context.

1.1.2 Process Models

Process models include a mathematical statement that depicts a process and a way to account for uncertainty about the process. To compose a process model we start by thinking about the true state (z) of an ecological system. That state could be the size of a population, the flux of nitrogen from the soils of a grassland, the number of invasive plants in a community, or the area of landscape annually disturbed by fire. We seek to understand influences on that state, the things that cause it to change. We write an equation,[5] a deterministic model that represents our ideas about the

[1]For elaboration, see Wikle (2003); Clark (2003b); Cressie et al. (2009), and Wikle et al. (2013).

[2]It might be useful to review scalars, vectors, and matrices. Most ecologists are familiar with matrices—rows and columns of numbers. Vectors are "one dimension" of a matrix, that is, a row or a column. Scalars are a single element. So, a matrix might be $\mathbf{A} = \begin{pmatrix} c & d \\ d & f \end{pmatrix}$; a vector, $\mathbf{a} = (c, d)'$; and a scalar, c. We will use the notation $()'$ to list the elements of a vector.

[3]We illustrate with the familiar example $y = \beta_0 + \beta_1 x$. The vector of parameters from the model is $\boldsymbol{\beta} = (\beta_0, \beta_1)'$. When discussing parameters generically, we will use $\boldsymbol{\theta}$.

[4]The symbol "|" reads conditional on.

[5]Or equations. To keep things simple, we focus on a single equation here, but we might build a system of equations making multiple predictions of states.

behavior of the state of interest and the quantities that influence it.[6] When we say the model is deterministic, we mean that for a given set of parameters and inputs, it will make precisely the same predictions. We use the notation $g(\boldsymbol{\theta}_p, \mathbf{x})$ to represent the deterministic part of a process model, where $g()$ is any mathematical function, $\boldsymbol{\theta}_p$ is a vector of parameters in the model, and $\mathbf{x}$ is one or more explanatory variables that we hypothesize influence the true state.

Our deterministic model is an abstraction, so it follows that we have omitted influences on the true state from the model, and we must deal with our omissions. If we model aboveground net primary production of a grassland as a function of growing season precipitation, we have brushed aside the influence of grazing intensity and precipitation that occurs during the dormant season. If we model reproductive success of individuals as a function of age and genotype, we have ignored variation contributed by their nutritional status. A model of harvest from a fishery based on observations of stock size and sea temperature omits the effect of variation in the food web. We recognize that these neglected influences shape the behavior of the true state by treating them stochastically, by estimating a parameter, σ_p^2, that subsumes all the unmodeled influences on the true state. Including this stochastic component allows us to estimate a statistical distribution (fig. 1.1.2A) for the true state,

$$\underbrace{\left[z|g\left(\boldsymbol{\theta}_p, \mathbf{x}\right), \sigma_p^2\right]}_{\text{process model}}, \tag{1.1.1}$$

where the bracket notation $[z|g(\boldsymbol{\theta}_p, \mathbf{x}), \sigma_p^2]$ means the distribution of z conditional on $g(\boldsymbol{\theta}_p, \mathbf{x})$ and σ_p^2.[7] If the notation is somewhat unfamiliar at this point, don't worry; equation 1.1.1 simply says that if we know the functional form $g()$ and the values of $\boldsymbol{\theta}_p$, $\mathbf{x}$, and σ_p^2, we can specify the probability distribution of the true state, z (fig. 1.1.2 A).

We want to determine the probability distribution of the true state as well as the probability distributions of the parameters in our model. Doing so

[6] As a tangible example, we might model the influence of phytoplankton biomass (x) on zooplankton biomass (z) with a linear model, $z = \theta_0 + \theta_1 x$. In this case $z = g(\boldsymbol{\theta}, x) = \theta_0 + \theta_1 x$.

[7] This notation was first introduced by Gelfand and Smith (1990) as a way to reduce clutter. It has become widely used by statisticians and ecologists. There is a small caveat needed here. We are using the arguments for probability distributions as if they were the mean ($g(\boldsymbol{\theta}_p, x)$) and variance ($\sigma_p^2$) of the distributions. The arguments for most of the statistical distributions are parameters that are functions of the mean and variance, a topic we treat in detail in section 3.4.4. For now, we take the liberty of treating the mean and variance as the needed parameters because they are familiar to ecologists.

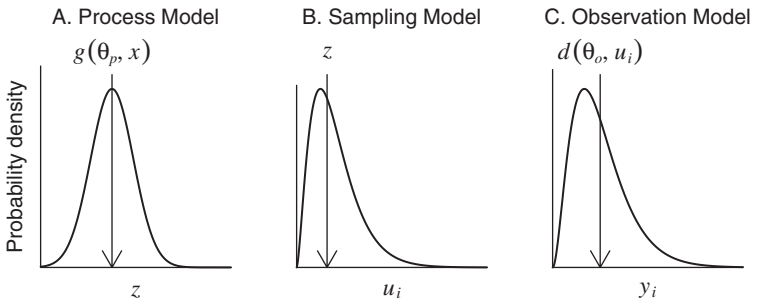


Figure 1.1.2. An observation on the operation of an ecological process (y_i) is linked to ideas about how the process works via three linked probability distributions. Hierarchical models consist of linked distributions. As we move from panel **A** to **C**, notice how the random variables (on the x-axis) becomes the the mean (or other central tendency) of the distribution in the next panel. **(A)** The deterministic model, $g(\boldsymbol{\theta}_p, \mathbf{x})$ predicts a true state, z, as a function of parameters $\boldsymbol{\theta}_p$ and predictor variables $\mathbf{x}$. There is uncertainty in the predictions, σ_p^2, that arises because there are influences on z that are not represented in the model. Thus, the distribution in **A** is $[z|g(\boldsymbol{\theta}_p, \mathbf{x}), \sigma_p^2]$. If the deterministic model predicts z well, then the distribution in **A** shrinks toward the $g(\boldsymbol{\theta}_p, \mathbf{x})$ arrow. **(B)** There are almost always more instances of z in nature than we can hope to observe. Observations of individual instances of z define a sampling distribution, $[u_i|z, \sigma_s^2]$, the breadth of which depends on σ_s^2. As variation among observations declines, the distribution in **B** shrinks toward the z arrow. Note that the mean of the distribution shown by the arrow is not at the peak of the distribution, because the distribution is skewed. (**C**) Observations (y_i) of instances of the true state u_i are often biased, such that $y_i \neq u_i$. A deterministic observation model, $d(\boldsymbol{\theta}_o, u_i)$, corrects for this bias. Uncertainty in this correction (σ_o^2) leads to the observation distribution $[y_i|d(\boldsymbol{\theta}_o, u_i), \sigma_o^2]$. As uncertainty in the observation model declines, the distribution in **C** shrinks toward the $d(\boldsymbol{\theta}_o, u_i)$ arrow.

requires evaluating the predictions of the process model against data. The data can be obtained in experiments or observational studies; they can be measurements we plan to collect or have already collected. This linkage between process models and observations is discussed next.

1.1.3 Sampling Models

We can rarely observe all instances of the true state in the system we study. Instead, we take a sample of $i = 1, \ldots, n$ observations of the true state and we notate the ith observation as u_i. This sample might be biomass from plots on a grassland landscape where we seek to understand the true state, aboveground productivity. It might be presence or absence of an exotic fungus on trees in a stand where we want to understand

infestation of the stand. It might be classifications of zooplankton in aliquots from a stream where we want to estimate the stream's species richness. Uncertainty arises because our sample assuredly will not represent the true state perfectly. Again, we represent this uncertainty stochastically using a probability distribution relating the true state to an observation

$$\underbrace{\left[u_i|z, \sigma_s^2\right]}_{\text{sampling model}}, \tag{1.1.2}$$

where σ_s^2 represents sampling variation (fig. 1.1.2 B). Expression 1.1.2 implicitly assumes that we can observe instances of the true state without bias, which simply says that if we collect many observations, then the average (i.e., also called the *expected value* of the observations, $\mathrm{E}(u)$) of the observations equals the mean of the distribution of the true state, $\mathrm{E}(u) = z$. (Expectation will be treated in detail in chapter 3.) We realize samples of the true state is a nuanced concept—soldier on, things will become clear in the next section.

1.1.4 Observation Models

The assumption that we can observe the true state perfectly may not be reasonable. When we count animals, some are overlooked. When we use Lidar[8] to estimate the heights of 10,000 trees, we do not measure the height of each tree using a ladder and a meter tape (thankfully) but instead observe backscatter from a laser beam. When we measure nitrogen mineralization, we do not follow the fate of individual nitrogen atoms but measure the net change in the extractable soil ammonium pool over time. The mismatch between what we observe and the true state requires a model of the observations, which we notate as $d(\boldsymbol{\theta}_o, u_i)$, where $\boldsymbol{\theta}_o$ are parameters. It is important to understand that u_i is the quantity we would observe if we could *perfectly* observe the instance of the true state in a draw from all the instances, without any bias injected by our observation process. We use y_i to notate the actual measurements we have in hand, including error resulting from the way we observe the u_i. The observation model serves to eliminate the bias found in our observations, y_i relative to an instance (u_i) of the true state drawn from the distribution of z. The probability distribution of the observations (fig. 1.1.2 C) arising from the

[8]Lidar, light detection and ranging, a technique used in remote sensing.

observed instances of the true state is

$$\underbrace{\left[y_i | d(\boldsymbol{\theta}_o, u_i), \sigma_o^2\right]}_{\text{observation model}}, \tag{1.1.3}$$

where the σ_o^2 represents all the influences on the y_i that are not represented in $d(\boldsymbol{\theta}_o, u_i)$. As a simple example, the σ_o^2 could be the variance of the predictions of a regression model used to calibrate observations against true values. We emphasize that model $d(\boldsymbol{\theta}_o, u_i)$ is needed to offset bias in the y_i. If our observations are unbiased, then there is no need for an observation model, (i.e., $y_i = u_i$) and the uncertainty in our data arises solely from sampling variation (i.e., eq. 1.1.3). For simplicity, we have ignored the sampling variation and observation errors that might influence the $\mathbf{x}$, but these could be handled in the same as we have done for the $\mathbf{y}$ (i.e., we would use probability distributions like eqs. 1.1.2 and 1.1.3).

1.1.5 Parameter Models

Because the approach we sketch is Bayesian, we also require models of the parameters expressing what we knew about the parameters when we began our investigation, that is, our *prior* knowledge. This knowledge is expressed in probability distributions, one for each parameter we seek to estimate[9]

$$\underbrace{\left[\boldsymbol{\theta}_p\right]\left[\boldsymbol{\theta}_o\right]\left[\sigma_p^2\right]\left[\sigma_s^2\right]\left[\sigma_o^2\right]}_{\text{parameter models}}. \tag{1.1.4}$$

These distributions must have numeric arguments that specify our current knowledge of the probability distribution of the parameters. The arguments can be chosen to make the distributions informative or vague, but as you will see, we will encourage you to make priors as informative as knowledge and scholarship allows. We might know a lot about a parameter or we might know very little.

1.1.6 The full Model

We are now equipped to write a mathematical expression representing our ideas about the operation of an ecological process linked to data in a way

[9]Equation 1.1.4 requires the assumption that the parameters $\boldsymbol{\theta}_p, \boldsymbol{\theta}_d, \sigma_p^2, \sigma_s^2$, and σ_o^2 are statistically independent. We will explain this idea in greater detail later.

that includes all sources of uncertainty—in the process, in our sample of the process, and in the way we observe it (fig. 1.1.2)

$$\left[\underbrace{z, \boldsymbol{\theta}_p, \boldsymbol{\theta}_o, \sigma_p^2, \sigma_s^2, \sigma_o^2, u_i}_{\text{unobserved}} \mid \underbrace{y_i}_{\text{observed}}\right] \propto$$

$$\underbrace{\left[y_i | d(\boldsymbol{\theta}_o, u_i), \sigma_o^2\right]}_{\text{observation model}} \underbrace{\left[u_i | z, \sigma_s^2\right]}_{\text{sampling model}} \underbrace{\left[z | g\left(\boldsymbol{\theta}_p, \mathbf{x}\right), \sigma_p^2\right]}_{\text{process model}} \underbrace{\left[\boldsymbol{\theta}_p\right]\left[\boldsymbol{\theta}_o\right]\left[\sigma_p^2\right]\left[\sigma_s^2\right]\left[\sigma_o^2\right]}_{\text{parameter models}}. \quad (1.1.5)$$

Equation 1.1.5 is Bayesian and hierarchical.[10] It is *Bayesian* because it treats the unobserved quantities as random variables. This treatment allows us to make statements about the probability distributions of all of the unobserved quantities based on the observed ones. It is *hierarchical* because the z and u_i are found on both sides of a conditioning symbol " | ", illustrating a powerful tool for simplifying problems that we will discuss more fully soon. The observation model includes our knowledge of the relationship between the true state and our observations of it and the uncertainty that occurs because that relationship is imperfect. The sampling model includes the uncertainty that comes from observing a subset of instances of the true state. The process model represents our hypotheses about the ecological process by specifying a probability distribution defining our knowledge and our uncertainty about the true state and the factors that control its behavior. The parameter models allow us to exploit previous estimates of parameters that we have made ourselves or that have been made by others. Together these models provide a line of inference extending from concepts to insight for a broad range of research problems (fig. 1.1.1). We can use equation 1.1.5 to obtain estimates of unobserved states, parameters, and quantities of interest derived from parameters and states. All of these estimates are properly tempered by uncertainty in a statistically coherent way.

In the remainder of this book, we develop the principles needed to understand equation 1.1.5 and to apply it to research problems in ecology. We tailor it to match the needs of the particular problem at hand. But first, we provide an example of its use.

[10]The symbol $\propto$ means "is proportional to." It may be confusing at this point that the observed **x** does not show up in the list of observed quantities on the left-hand side of the proportionality. Bear with us. We will make that clear as we proceed.

1.2 An Example Hierarchical Model

We are now going to apply the general framework we have previewed to a specific problem. Remember that we don't expect this section to be familiar to you. Although the application here focuses on learning about population dynamics of a large mammal from a time series of observations in East Africa, it could be about any topic, any research design, and any location. We urge you to draw analogies to your own work as the example develops.

The Serengeti wildebeest (*Connochaetes taurinus*) population migrates across the grasslands of Tanzania and Kenya in an annual cycle driven by availability of green plant biomass (Boone et al., 2006). During the late 1800s, the viral disease rinderpest created a panzootic, decimating the wildebeest and other wild and domestic ruminants. Numbers of these animals remained low until the 1950s when a campaign to vaccinate cattle in the pastoral lands surrounding the Serengeti eliminated the virus in wildlife (Plowright, 1982). Survivorship of wildebeest, particularly juveniles, doubled after rinderpest was eradicated, and the population grew rapidly until the mid 1970s when density-dependent mortality produced a quasi-equilibrium. This steady state appears to have been caused by intraspecific competition for green plant biomass during the dry season. This biomass varies in approximate proportion to annually variable rainfall (Mduma et al., 1999).

Developing informative models depends on a clearly stated question. An unambiguous question is vital because it guides the abstraction that we formulate; it informs our decisions about which variables and parameters we need to include in our model. In this particular example we ask the question: How does variation in weather modify feedbacks between population density and population growth rate in a population of large herbivores occupying a landscape where precipitation is variable in time? Answering this question requires a model that portrays density dependence, effects of precipitation, and their interaction.

1.2.1 Process Model

We pose the deterministic model portraying growth of the wildebeest population as

$$N_t = g(\boldsymbol{\beta}, N_{t-1}, x_t, \Delta t) = N_{t-1} e^{(\beta_0 + \beta_1 N_{t-1} + \beta_2 x_t + \beta_3 N_{t-1} x_t)\Delta t}, \qquad (1.2.1)$$

where N_t is the unobserved, true abundance of wildebeest in year t (corresponding to z in eq. 1.1.5); x_t is a measure of the annual rainfall

that influences population growth during $t - 1$ to t; and $\Delta t = 1$ year.[11] The meaning of the coefficients $\boldsymbol{\beta}$ requires some thought. A critical part of gaining insight from models and data is careful thinking about the biological meaning of the parameters in our models and their relationship to existing concepts and theory, the need for which is illustrated here.

We first consider the intercept, β_0, a logical place to start. If the units of precipitation, x_t, are centimeters per year, ranging from 0 to a large number, then algebra dictates that $e^{\beta_0 \Delta t}$ is the proportional change in the population during the period t to $t + 1$ that occurs when abundance is zero and rainfall is zero. The abundance equals zero part might seem a bit odd at first glance, because there is no population to grow when $N_t = 0$, but the parameter nonetheless serves to define the upper limit on population growth rate as numbers approach zero. There is no problem here. But the part about zero rainfall is troubling because it makes β_0 difficult to interpret biologically or at least makes it somewhat unuseful in its biological interpretation. Why would we want to estimate a parameter determining population growth when the population is low and rainfall is low? Alternatively, if we define x_t in terms of divergence from the long-term average by subtracting the mean rainfall from each year's observation, then the definition of β_0 becomes more sensible. In this case, $e^{\beta_0 \Delta t}$ is the proportional change in population size that occurs when the population is zero and rainfall is average. This definition allows us to relate our model to well-established theory; β_0 is analogous to the intrinsic rate of increase in the logistic equation.[12]

Once we have defined the intercept in a sensible way, the slope terms are easily interpreted. The parameter β_1 represents the strength of density dependence, that is, the change in the per capita population growth rate that occurs for each animal added to the population.[13] Again relating our model to classical theory, $\beta_1 = -(r_{max}/K)$, where K is the carrying capacity, that

[11] The identical form $\log(N_t) = \log(N_{t-1}) + (\beta_0 + \beta_1 N_{t-1} + \beta_2 x_t + \beta_3 N_{t-1} x_t)\Delta t$ might be more familiar to ecologists accustomed to a general linear modeling framework.

[12] This might require a bit of explanation. Equation 1.2.1 can be rearranged to $\log(N_t/N_{t-1}) = (\beta_0 + \beta_1 N_{t-1} + \beta_2 x_t + \beta_3 N_{t-1} x_t)\Delta t$. If we drop the $\beta_2 x_t$ and $\beta_3 x_t N_{t-1}$ terms and assume Δt is small, the resulting expression approximates $1/N \cdot dN/dt = r - rN/K$, where N is the population size, r is the intrinsic rate of increase, and K is the carrying capacity—the long-term average population size at which $1/N \cdot dN/dt = 0$. Thus, the portion of our model representing density dependence is a version of the logistic equation in discrete time, also known as the *Ricker equation*, where $\beta_0 = r$, and $\beta_1 = r/K$.

[13] You might be tempted to make the sign for β_1 negative, because density dependence reduces the per capita population growth rate as population size increases. There is a good reason to use addition. When the model is fit to data, we want to estimate the sign and the value of β_1 without needing to "back transform" the sign. Always use addition in additive models that you will fit to data.

is, the population size at which the long-term average population growth rate equals zero. The parameter β_2 expresses the strength of the effect of variation in rainfall, that is, the change in the per capita population growth rate per unit deviation from the long-term mean. Finally, the parameter β_3 determines the magnitude of the effect of rainfall on the effect of density.

Estimating the β's will inform the question we posed, but we don't pretend that they capture all the influences on wildebeest population dynamics. We recognize that there are other processes, for example, predation, poaching, and disease, that shape wildebeest numbers over time. We can acknowledge these other processes exist without portraying them explicitly. Instead, we lump these unmodeled effects into a single parameter, σ_p^2. The model of the process for a population at time t including deterministic and stochastic components is

$$\left[N_t | g\left(\boldsymbol{\beta}, N_{t-1}, x_t, \Delta t\right), \sigma_p^2\right]. \tag{1.2.2}$$

1.2.2 Sampling Model

The wildebeest population was estimated[14] on 20 occasions during 1961–2008 using spatially replicated counts of animals on georectified aerial photographs arrayed along transects, with each photograph covering a known area (Norton-Griffiths, 1973). For simplicity, we ignore the aspect of sampling contributed by the transects and treat the photographs as if they were a random sample from the area used by the population. Thus, we assume that each photograph provided a statistically independent estimate of population density calculated as the observed count divided by the area covered by the photograph. The stochastic sampling model representing the relationship between these observations and the true population size is

$$\left[y_{tj} \,\middle|\, \frac{N_t}{a}, \sigma_s^2\right], \tag{1.2.3}$$

where y_{tj} is the density of animals at year t on photograph j, and a is the total area from which the sample of photographs was drawn—the area used by the population we are modeling—which we assume to be a known constant. Note that equation 1.2.3 implies that there is no bias in the estimate of animal density on a given photograph, which is the assumption made

[14]Portions of this time series have been published in Hilborn and Mangel (1997) and Mduma et al. (1999). The most recent data were graciously provided by A.R.E. Sinclair, Ray Hilborn, and Grant Hopcraft. We use these data later to illustrate making inference from a single model.

by the researchers who collected the data. Although this assumption is reasonable for large animals counted in open habitat, we could obviate the need for this assumption by modeling the animals that were present and not observed.

As a purely hypothetical example, imagine that we fitted a sample of animals with high-resolution telemetry instruments that would allow us to know whether each animal was within a photograph. Imagine also that we marked them with highly visible neckbands to allow us to distinguish between animals that were fitted with instruments and those that were not. We could use these observations to estimate the probability (ψ) that an animal truly present was counted. In this case our combined observation and sampling model would be

$$\underbrace{\left[y_{tj} | \psi n_{tj}, \sigma_o^2\right]}_{\text{observation model}} \underbrace{\left[n_{tj} \,\middle|\, \frac{N_t}{a}, \sigma_s^2\right]}_{\text{sampling model}}, \tag{1.2.4}$$

where n_{tj} are animals that are truly present on photograph j during year t, and σ_o^2 is the uncertainty associated with the estimate of ψ. We could also deal explicitly with the likely cases in which it was not certain whether an animal was on a sampled photograph.

To simplify the example and to exploit published data, we make the heroic assumption that rainfall at time t, x_t, is measured without error; we are treating it as known, but we are not obliged to do so. We could develop sampling models and observation models for the rainfall data in the same way we did for the count data (i.e., eqs. 1.2.3 and 1.2.4).

1.2.3 Full Model

Combining the sampling model (eq. 1.2.3) and process model (eq. 1.2.2) with models for the parameters, we obtain the foundation for a full Bayesian analysis capable of estimating the parameters and the unobserved, true population size (fig. 1.2.1); however, work remains to be done. The full model written thus far (fig. 1.2.1) applies to a single year of observations and a single photograph, but, of course, we want to use all the years and all the observations within a year. We have not yet chosen specific probability distributions for the stochastic components, and we must do so in a sensible way. We may need to deal with correlation among photographs or among annual estimates in the time series of data, and with errors in the estimation of rainfall. We need to lay out a method for numerically estimating the parameters and unobserved states. However, although tasks remain, all

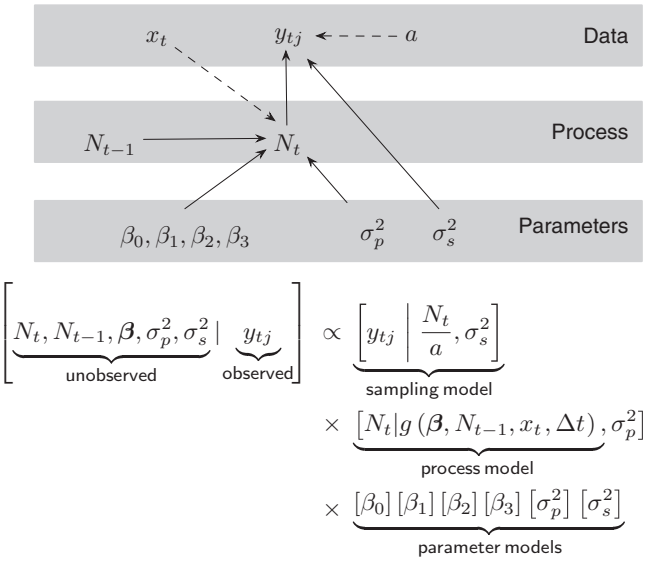


Figure 1.2.1. Hierarchical Bayesian model of the dynamics of the Serengeti wildebeest population. The true population size at time t is modeled using the deterministic model $g(\boldsymbol{\beta}, N_{t-1}, x_t, \Delta t) = N_{t-1} e^{(\beta_0 + \beta_1 N_{t-1} + \beta_2 x_t + \beta_3 N_{t-1} x_t)\Delta t}$, which represents the effects of the true, unobserved population size (N_t); the dry season rainfall (x_t); and their interaction on population growth. The biological interpretation of the parameters $\boldsymbol{\beta}$ are given in the text. The parameter σ_p^2 represents all the effects on wildebeest abundance not represented by the $\boldsymbol{\beta}$. The y_{tj} is a single observation of population density obtained from one of $j = 1, \ldots, n_t$ photographs chosen from area $= a$ km^2. The parameter σ_s^2 represents sampling error. The arrow diagram is a Bayesian network, also called a *directed acyclic graph*, that can be used as a visual guide for properly writing the full model. The solid lines show stochastic relationships. The dashed lines show deterministic relationships, implying that the quantities at the tails of the arrows are known without error. Instructions for drawing diagrams like this one will be developed in subsequent chapters.

proceed in a logical way from the foundation we have built (fig. 1.2.1). All are manageable. Completing them allows us to reliably estimate unobserved states, parameters, and derived quantities of interest.

1.3 What Lies Ahead?

One of the aims of this book is to enable ecologists to write equations for models that allow data to speak informatively. Our goal is to provide

an understanding of principles needed to accomplish this vital task. We will show how an enormous range of research problems in ecology can be decomposed into a set of sensible parts, as we have illustrated. The specific example we offered no doubt raised more questions than it answered, which was our intention. Notable among these questions might be the following:

1. Why does the Bayesian approach to analysis work? What are the probabilistic foundations of inference from models like this one?
2. How does a Bayesian analysis relate to analyses based on maximum likelihood?
3. How do we choose the appropriate statistical distributions for the stochastic components of the model (fig. 1.2.1)?
4. How can we incorporate multiple sources of data in estimates of parameters and states?
5. How do numerical methods allow us to implement the model and obtain results? How do these methods work?
6. How can we evaluate the model to assure it adequately represents the data?
7. What can we conclude about the operation of the ecological process based on estimates of the parameters and states? What do we do about derived quantities; for example, how do we make inferences about the population size where growth rate is maximum?
8. What if we want to evaluate the strength of evidence for this model (equation 1.2.1) relative to competing ones, for example, a model with nonlinear density dependence?
9. How do we predict future states with honest estimates of uncertainty?

Answering these questions in a clear and accessible way for a broad range of research problems in ecology is the goal of remainder of this book.

2 Deterministic Models

We write models to express our ideas about how ecological processes work (figure 0.0.1 A). We then use data to decide if those ideas are any good. Combining data with models provides insight about ecology by answering a fundamental question: What is the probability that I will observe the data if the model faithfully represents the processes that give rise to the data? Our starting point for answering this question will always be consideration of an ecological process. We formalize our thoughts by writing a deterministic equation or equations making predictions that can be compared with observations (fig. 0.0.1 A). There is no uncertainty in our predictions because the model is deterministic. In this chapter we talk about deterministic models as expressions of ecological hypotheses. In the next chapter we lay out the principles of probability that allow us to combine these models with data in a way that provides new knowledge qualified by uncertainty.

2.1 Modeling Styles in Ecology

We wrote this book to be broadly useful across all the subdisciplines of ecology, offering a way of working with models and data that is recognizably valuable, whether you study genes or ecosystems. To that end, we briefly discuss the different ways that deterministic models have been used in ecology. We want to ensure that ecologists who work on different kinds of problems recognize a familiar style of modeling, and have confidence that what lies ahead is relevant to their own research.

Modeling has formed a productive, diverse activity throughout the history of ecology (Kingsland, 1985; Golley, 1993). Subdisciplines of

ecology have developed different modeling "cultures," as is plainly seen in textbooks about quantitative ecology, all of which teach how to use models of one type or another. Despite their ostensibly shared focus on modeling, it is striking how starkly *different* these books are; some lack a single topic in common in their tables of contents (e.g., compare Schwartzman and Kaluzny, 1987; Gotelli and Ellison, 2004; Otto and Day, 2007; Railsback and Grimm, 2012).

The marked divergence among these texts reflects fundamental differences among modeling traditions in ecology. We identify three main ones, which we call theoretical, empirical, and simulation (table 2.1). We do not argue that these modeling styles are distinct; of course, there is overlap among them. Nonetheless, categorizing them this way offers a useful heuristic that helps us organize our thinking about how models have been used in the past and how they might be used more fruitfully in the future if each style could exploit the strengths of the others.

2.1.1 Theoretical

Seminal ideas in ecology were motivated by mathematical analysis of systems of deterministic, nonlinear differential and difference equations (Levins, 1969; May, 1977; Anderson and May, 1979; MacArthur and Wilson, 2001). No data are required in this modeling style—insight comes directly from mathematical properties of the equations forming the model. Discovering equilibrium points and conditions for local stability—the workhorses of ecological theoreticians—are examples of this type of insight (Edelstein-Keshet, 1988; Otto and Day, 2007). The absence of data from these models and their simplicity are what allows them to be used to generalize about the entire range of possibilities for ecological states. When data are used to specify values for parameters, the rich range of behavior of these models is abruptly compressed to the specific behavior dictated by such data (Levins, 1966). The theoretical style depends on simple models, because nonlinear systems with many state variables[1] and parameters are mathematically intractable.

The great strength of the theoretical tradition in ecological modeling is that the parameters and state variables are defined in biological terms. They symbolize quantities in nature, and the model symbolically portrays how a process works. These models rigorously express ideas about *why*

[1] *State variables* are the quantities of interest in models that change over time or space. Parameters describe how those changes occur. For example, N is a state variable in the logistic equation, $dN/dt = rN(1 - N/K)$, and r and K are parameters.

TABLE 2.1
Comparison of Prevailing Modeling Styles in Ecology

	Theoretical	*Empirical*	*Simulation*
Number of variables	few	few	many
Requires data for insight?	no	yes	yes
Fields in which most often applied	theoretical ecology	all subdisciplines of ecology	ecosystem ecology, landscape ecology
Definition of parameters	as biological quantities	as relationships in data	as biological quantities
Basis for uncertainty	NA*	probability theory	ad hoc
Example texts	Edelstein-Keshet (1988); Otto and Day (2007); Diekman et al. (2012)	Sokal and Rohlf (1995); Gotelli and Ellison (2004)	Schwartzman and Kaluzny (1987); Railsback and Grimm (2012)

* Uncertainty refers to things that are unknown. Theoretical models may use stochastic, simulation, but they do not distinguish between quantities that are known and those that are unknown.

outcomes occur by abstracting the mechanism that produces them. Thus, the theoretical style is a particularly valuable way to develop hypotheses, which, in turn, are often tested using empirical models.

2.1.2 Empirical

Empirical models describe relationships in data. The preponderance of modeling in ecology has used this style in regression, analysis of variance, and related analyses. All these methods depend on a deterministic model at their core, but many ecological researchers rarely give these models much thought—linear models are a ubiquitous default. The parameters in these models customarily lack portable, biological interpretation; most are defined solely in terms of the data used to estimate them. Nonetheless, purely empirical models describing patterns in data— that is, models with no biological definition for their parameters—have motivated theoretical models that represent the mechanisms producing the patterns (Spalinger and Hobbs, 1992; Ritchie, 1998; Brown et al., 2002), forming a revealing interplay between modeling styles in ecology.

The great strength of the empirical style of modeling is its foundation in statistics, a foundation that allows rigorous statements about uncertainty. The close connection between empirical models and agronomic-style experimental designs means that these models are usually quite simple. Simplicity allows isolation of the influence of a few variables on an ecological state. However, the simplicity of empirical models means they cannot represent multiple interactions and the effects of composites of many variables. The need to represent the interaction of multiple states motivated the use of simulation models.

2.1.3 Simulation

The computing power of the entire U.S. Department of Defense in 1970 was less than that of a smartphone today. Before the arrival of fast, cheap computers, the theoretical and empirical styles of ecological modeling were the only modeling alternatives. A new, third style of modeling based on computer simulation emerged in the mid-1970s with the publication of the first model representing the dynamics of an entire ecosystem (Innis, 1978). The original simulation models did not differ qualitatively from the systems of differential equations that predominated in the theoretical tradition. As in the theoretical style, parameters and state variables included in the simulation approach enjoyed clear biological definitions. The important difference was in dimensionality—the number of state variables included—and as a result, in the analysis of the models. Gaining insight from nonlinear models with many state variables required assigning specific values to parameters and using computer simulation to visualize the change in state variables over time. The simulation approach has expanded dramatically since the 1970s to include various types of spatiotemporal models that exploit geographic information systems. A more recent development is the modeling of interactions among individuals (Judson, 1994; Railsback and Grimm, 2012).

It is often said that ecology is foremost the study of interactions. The strength of the simulation approach is its ability to represent multiple interactions extending across spatial scales and levels of organization, interactions that cannot be included in traditional statistical or mathematical analysis. Including these interactions is possible because simulation models aggregate multiple relationships among variables, relationships that are often developed using empirical and theoretical approaches. These models do not flinch at complexity, dispensing with the reductionism of the empirical and theoretical modeling styles. The primary weaknesses of the simulation approach are the absence of a statistically coherent way to deal with uncertainty and a somewhat cultural tendency to build models of unbridled

complexity. Until recently, the problem of estimating uncertainty has been treated in a largely ad hoc fashion, without a principled, probabilistic foundation (but see Clark et al., 2011; Luo et al., 2011; Gudimov et al., 2012; Araujo et al., 2013; Braakhekke et al., 2013; Broquet et al., 2013; Fiechter et al., 2013; Pacella et al., 2013; Wikle et al., 2013).

2.1.4 What Is Needed?

The prevailing modeling styles in ecology have unique strengths: the theoretical approach has encouraged ecologists to write equations portraying how processes work; the simulation approach allows us to represent multiple interactions and dependencies; and the empirical approach reliably includes uncertainty. What is needed is a way to merge styles in a way that exploits the strength of each. The Bayesian hierarchical framework encourages that merger by admitting deterministic models of all types, by allowing use of multiple sources of data, and by providing a broadly useful framework for quantifying uncertainty in multidimensional and complex models. A key feature of this book is that it teaches how to incrementally add complexity to simple ecological models so that uncertainty is properly modeled at each increment in model development. This incremental approach to model building requires a set of building blocks—deterministic models to represent our ideas about the operation of processes, which we treat in this chapter, and probability models to get the uncertainties right, which we describe in the following chapter.

2.2 A Few Good Functions

For the remainder of this chapter we outline a few deterministic models widely used in ecology. We do not attempt a comprehensive treatment, because other sources cover this topic lucidly and in depth (Otto and Day, 2007; Bolker, 2008). However, we will develop familiarity with a set of core functions that are widely used to represent ecological process across all subdisciplines and that will reappear throughout the book, particularly in chapters 6 and 12.

In all areas of ecology we seek to understand processes. We cannot escape the use of models to achieve that understanding; even the empirical tradition depends on them. A deterministic model of an ecological process is a function, $g(\,)$, that returns an output μ representing a true state z based on parameters ($\boldsymbol{\theta}_p$) and observations ($\mathbf{x}$):

$$\mu = g\left(\boldsymbol{\theta}_p, \mathbf{x}\right). \tag{2.2.1}$$

We distinguish the deterministic representation of the true state (μ) and its true value (z) because many influences that shape the behavior of the true state are not included in our deterministic model, as we will soon describe (chapter 3). The range of inputs to the function is called the *domain*, and the range of outputs is called the *range*. For simplicity, we assume the prediction of our model can be represented as a scalar, but it could be a vector, a matrix, or even a multidimensional array. We will use the function $g(\,)$ to represent a deterministic model throughout the book, but the specific symbols chosen for arguments will vary. When more than one deterministic model is needed, we will call $h(\,)$ into service.

2.2.1 Functions for Additive Effects

2.2.1.1 Simple Linear

The linear model

$$g(\boldsymbol{\beta}, \mathbf{x}) = \beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n \tag{2.2.2}$$

and its generalizations are familiar tools in ecological modeling, particularly in the empirical tradition. The model is widely useful because it expresses effects that are additive, providing a simple way to represent the relationship between predictor variables[2] (the x-values) and a response. Interactions can easily be represented with product terms, and curvilinear effects can be portrayed by raising the x to powers, operations that should be familiar from training in basic statistics. We usually start with the additive, linear model if we lack a solid understanding of mechanisms.

We will soon learn that deterministic functions must have an appropriate range for their predictions to be linked with data in a statistically reliable way. For example, the range of equation 2.2.2 is all real numbers, so it would be fine for describing the change in biomass of a forest during an interval of time, because "change in biomass" is a continuous quantity that can be positive or negative. But it would not be strictly correct for describing the stand biomass at any point in time, because biomass cannot be negative. It would also not be appropriate for describing the proportion of a landscape that is burned, a quantity limited to values between 0 and 1. Equations must provide predictions matched to the ecological quantities we wish to understand.[3]

[2] Also called *covariates*.

[3] More formally, predictions must have the proper *support*, a concept we describe fully in the next chapter.

2.2.1.2 Exponential

We can model quantities that are nonnegative while preserving the additive structure of the simple linear model using

$$g(\boldsymbol{\beta}, \mathbf{x}) = \exp(\beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n), \tag{2.2.3}$$

which maps the negative values of equation 2.2.2 to $(0, 1)$ and maps the nonnegative values to $(1, \infty)$ (fig. 2.2.1 A,B). You may have seen this model written as

$$\log(\mu) = \beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n, \tag{2.2.4}$$

using an algebraic rearrangement of equation 2.2.3. This version is widely used in the general linear modeling tradition, where $\log(\mu)$ is called a *link* function, in this case, a log link. When authors state they have log-transformed the response variable for a linear regression or are using a linear model with a log link, they are effectively using an exponential model (eq. 2.2.3).[4] We will use the untransformed version (eq. 2.2.3) in later examples because linear models are to be considered as one of many alternatives for functions used to represent ecological processes—those that are part of the general linear family, as well as those that cannot be transformed to a linear form.

2.2.1.3 Inverse Logit

Many quantities of interest in ecology are proportions—quantities that take on values ranging from 0 to 1. These can be modeled deterministically using the inverse logit function[5] (fig. 2.2.1 C,D),

$$g(\boldsymbol{\beta}, \mathbf{x}) = \frac{\exp(\beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n)}{1 + \exp(\beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n)}. \tag{2.2.5}$$

Equation 2.2.5 returns a real number less than or equal to 1 and greater than or equal to 0. As its name implies, the function is the inverse of the logit function, which, for a proportion p, is defined as

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right). \tag{2.2.6}$$

[4] Be sure you understand that this is the natural log. Statisticians almost never use the notation $\ln(\,)$, preferring $\log(\,)$ to mean the same thing.

[5] Also called the *logistic function*.

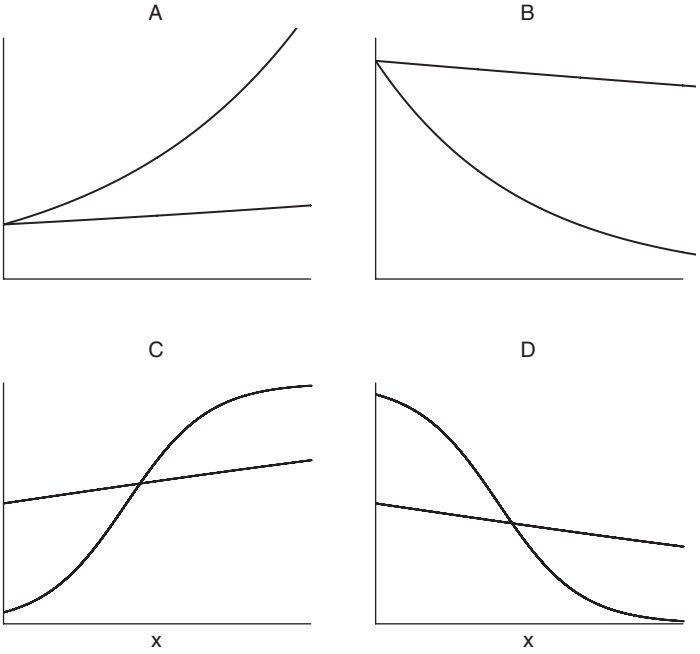


Figure 2.2.1. Examples of an exponential model for additive relationships, $\mu = \exp(\beta_0 + \beta_1 x)$, with a positive (**A**) and negative (**B**) slope. Examples of the inverse logistic model for additive relationships, $\mu = \frac{\exp(\beta_0+\beta_1 x)}{1+\exp(\beta_0+\beta_1 x)}$, with a positive (**C**) and a negative (**D**) slope. In each of the panels, the different curves show how varying the parameter β_1 controls the "straightness" and slope of the curve. Larger values of β_1 cause greater curvilinearity. Smaller values straighten the curve.

Solving for p in

$$\text{logit}(p) = \beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n, \tag{2.2.7}$$

we obtain equation 2.2.5. The logit link function used in generalized linear modeling is equation 2.2.7. If we multiply equation 2.2.5 by a, it returns values between 0 and a maximum a,

$$g(\boldsymbol{\beta}, a, \mathbf{x}) = \frac{a \exp(\beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n)}{1 + \exp(\beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n)}. \tag{2.2.8}$$

Before we move to a second set of functions, it is worth reemphasizing a point we made in the preview (sec. 1.2), namely, it is best practice to compose all the preceding linear models using addition operators even if

subtraction might seem to make greater sense biologically. This is because we want the *data* to determine the value of each of the coefficients in the models we fit, including the sign of the coefficient. Using subtraction can lead to confusion and errors, because then we must remember that a negative coefficient for a subtracted value means the sign of the coefficient is actually positive. If we use addition throughout, then the signs of the coefficients are directly interpretable without mental gymnastics.

2.2.1.4 Are These Really "Linear" Models?

When we introduce these models to students, they often respond, "These models are not really linear; both produce "curves" when I plot them. What do I do if I want to fit a straight line to strictly positive data or to data that range between 0 and 1?" The answer, of course, is that both functions can yield results that are "straight"[6] or curved depending on the values of the coefficients (fig. 2.2.1). Thus, we really have the best of both worlds – the exponential and inverse logit functions can have substantial curvature if we need nonlinearity, or can be straight if we don't.

2.2.2 Power Functions

The power function has seen broad application in ecology as a way to describe scaling, that is, how phenomena change as the dimensions of time, space, or mass change. There is an enormous amount of literature on this topic (reviewed by Peters, 1983; Pennycuick, 1992; Schneider, 1997; West et al., 2003; Marquet et al., 2005; Xiao et al., 2011). The power function is

$$g(\boldsymbol{\beta}, x) = \beta_0 x^{\beta_1} \tag{2.2.9}$$

and is often transformed to a linear form. Recall that the untransformed model predicts μ, so a linearized version is

$$\log(g(\boldsymbol{\beta}, x)) = \log(\mu) = \log(\beta_0) + \beta_1 \log(x)\,. \tag{2.2.10}$$

[6]We put *straight* in quotation marks because the functions do not have constant slope. However, the changes in slope can be sufficiently minor that the model provides a fine approximation of a function that truly has a constant slope, as does $g(\boldsymbol{\beta}, x) = \beta_0 + \beta_1 x$.

It is possible to include multiple independent variables (**x**) in power functions[7] using

$$g(\boldsymbol{\beta}, \mathbf{x}) = \beta_0 x_1^{\beta_1} x_2^{\beta_2} \ldots x_n^{\beta_n}, \tag{2.2.11}$$

or in linear form,

$$\log(\mu) = \log(\beta_0) + \beta_1 \log(x_1) + \beta_2 \log(x_2) + \ldots + \beta_n \log(x_n). \tag{2.2.12}$$

2.2.3 Asymptotic functions

Asymptotic processes are ubiquitous in ecology and for good reason—there are no quantities in biology that can increase or decline without limit. Asymptotes occur in models where the rate of change in the dependent variable slows with increases in the independent variable. These relationships are represented by a family of models, the best known of which, perhaps, is the Michaelis—Menten equation. It was originally formulated as a model of enzyme kinetics,

$$g(v_{max}, k, x) = \frac{v_{max} x}{k + x}, \tag{2.2.13}$$

where v_{max} is the maximum value of the independent variable (the true state, μ), approached asymptotically as x approaches infinity, and k is the half-saturation constant, the value of x where μ is half of its maximum. A commonly used rearrangement with parameters that are somewhat more interpretable biologically is

$$g(v_{max}, \alpha, x) = \frac{v_{max} x}{\frac{v_{max}}{\alpha} + x}, \tag{2.2.14}$$

where the parameter α is defined as the rate of change in μ at low values of x. Equation 2.2.13 has an intercept at $x = 0, \mu = 0$. This can easily be modified by adding a constant to x:

$$g(v_{max}, \alpha, c, x) = \frac{v_{max}(x + c)}{\frac{v_{max}}{\alpha} + (x + c)}. \tag{2.2.15}$$

[7] You will also see these written as $g(\boldsymbol{\beta}, \mathbf{x}) = \exp(\beta_0) x_1^{\beta_1} x_2^{\beta_2} \ldots x_n^{\beta_n}$ and $\log(\mu) = \beta_0 + \beta_1 \log(x_1) + \beta_2 \log(x_2) + \ldots + \beta_n \log(x_n)$.

If c is positive, the curve shifts left; if c is negative, the curve shifts right. Exponentiation of x and k in equation 2.2.13 creates the Hill function,[8]

$$g\left(v_{max}, k, q, x\right) = \frac{v_{max} x^q}{k^q + x^q}, \tag{2.2.16}$$

which represents sigmoidal processes with positive slopes when q is positive or negative slopes when q is negative. The magnitude of q controls the steepness of the slope.

The Michaelis–Menten equation is related to another commonly used asymptotic function, best known to ecologists as the model of Type II functional response first described by C. S. Holling (1959). If we substitute $v_{max} = 1/\gamma$ and $k = 1/\alpha\gamma$ in equation 2.2.13, we obtain

$$g\left(\alpha, \gamma, x\right) = \frac{\alpha x}{1 + \alpha\gamma x}. \tag{2.2.17}$$

Other asymptotic functions widely used in ecological models are the monomolecular function,

$$g\left(\alpha, \gamma, x\right) = \alpha\left(1 - e^{-\gamma x}\right), \tag{2.2.18}$$

and the Gompertz function,

$$g\left(\alpha, \gamma, x\right) = e^{-\alpha e^{-\gamma x}}. \tag{2.2.19}$$

2.2.4 Thresholds

Identifying thresholds[9] in ecological processes has emerged as an especially important challenge for ecology in a world experiencing rapid and manifold changes in physical and biological processes. The essential problem is to fit functions that take the form

$$\mu = \begin{cases} g_1\left(\boldsymbol{\theta}_1, x\right) & x < \tau \\ g_2\left(\boldsymbol{\theta}_2, x\right) & x \geq \tau \end{cases}, \tag{2.2.20}$$

where τ is a change point—a parameter we estimate that controls an abrupt change in the process. The change in the process is portrayed by

[8] A. V. Hill (1910) developed this model to represent oxygen binding to hemoglobin.

[9] Also known by the trendy term *tipping points*.

a shift between the two models, $g_1(\boldsymbol{\theta}_1, x)$ and $g_2(\boldsymbol{\theta}_2, x)$, which can be any deterministic function including all the ones we have discussed so far in this chapter. The challenge is to estimate $\boldsymbol{\theta}_1$ and $\boldsymbol{\theta}_2$. We point this out because Bayesian model fitting using the Markov chain Monte Carlo algorithm (chapter 7) allows us to accomplish this estimation in cases where maximum likelihood approaches have historically been defeated by the technical requirement for smooth functions with continuous derivatives. Abrupt changes like those depicted in equation 2.2.20 have been very difficult to fit using methods that require maximization or minimization using numerical methods, but they can easily be estimated using the approaches we describe later in the book.